

An integrated method for targeted Oxford Nanopore sequencing and automated bioinformatics for the simultaneous detection of bacteria, fungi, and ARG

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Abstract

Aims: The use of metagenomics for pathogen identification in clinical practice has been limited. Here we describe a workflow to encourage the clinical utility and potential of NGS for the screening of bacteria, fungi, and antimicrobial resistance genes (ARGs).

Methods and results: The method includes target enrichment, long-read sequencing, and automated bioinformatics. Evaluation of several tools and databases was undertaken across standard organisms ($n = 12$), clinical isolates ($n = 114$), and blood samples from patients with suspected bloodstream infections ($n = 33$). The strategy used could offset the presence of host background DNA, error rates of long-read sequencing, and provide accurate and reproducible detection of pathogens. Eleven targets could be successfully tested in a single assay. Organisms could be confidently identified considering $\geq 60\%$ of best hits of a BLAST-based threshold of e -value 0.001 and a percent identity of $>80\%$. For ARGs, reads with percent identity of $>90\%$ and $>60\%$ overlap of the complete gene could be confidently annotated. A kappa of 0.83 was observed compared to standard diagnostic methods. Thus, a workflow for the direct-from-sample, on-site sequencing combined with automated genomics was demonstrated to be reproducible.

Conclusion: NGS-based technologies overcome several limitations of current day diagnostics. Highly sensitive and comprehensive methods of pathogen screening are the need of the hour. We developed a framework for reliable, on-site, screening of pathogens.

Impact Statement

The use of next-generation sequencing (NGS) in clinical environments can better describe pathogen profiles and reduce the need for broad spectrum empirical therapy. We observe that NGS-based screening can be a viable option as the primary screening tool in clinical settings.

Keywords: infectious disease; genomics; next-generation sequencing; nanopore sequencing; antibiotic resistance gene; bioinformatics; clinical genomics

Introduction

The Koch's postulate states that the proof of an infection is in our ability to detect and isolate the organism and prove its causality in disease. For more than a century, microbiological and biochemical investigations have formed the bedrock of confirming clinical suspicion of infection. The shift to nucleic acid-based characterization of microorganisms led to a re-evaluation and rephrasing of Koch's postulates (Hosainzadegan et al. 2020, Kobras et al. 2021). While PCR-based methods have revolutionized the targeted, rapid detection of pathogens, DNA sequencing has fundamentally altered the definitions of microbiomes and the scope of pathogens in clinical samples (Pechorsky et al. 2009, Sanghavi et al. 2012, Chiu and Miller 2019, Church et al. 2020, Fida et al. 2021). Several sequencing strategies spanning metagenomics, whole genome metagenomics and metabarcoding have been developed to el-

evate genomics-based microbiome profiling to clinical use. Based on the clinical requirements, an ideal solution for the diagnosis of infections in a healthcare facility needs to fulfil three conditions: (i) low turn around time (TAT); (ii) comprehensive; and (iii) low cost (Rossen et al. 2018, Gu et al. 2019, Fida et al. 2021, Kobras et al. 2021).

Early application of PCR amplified targets such as the rRNA genes (16S, 18S, 23S) combined with Sanger sequencing established the role of genomics in the routine identification of bacteria and fungi across laboratories (Church et al. 2020). Next-generation sequencing (NGS) overcame the time-consuming and sequential sequencing approach of first-generation sequencing (Gu et al. 2019, Koonin et al. 2021). The increased throughput of NGS in combination with the relevant bioinformatic tools enabled querying whole genomes or directly analysing entire communities using metagenomics

with high accuracy. One of the key aspects of the utility of NGS is our ability to tune the resolution of taxonomy depending on the application (Govender et al. 2021, Kobras et al. 2021, Deng et al. 2022, Andres et al. 2023). One of the conflicts in the choice of sequencing technology is in the use of short-read sequencing (SRS) such as Illumina and long-read sequencing (LRS) such as Oxford Nanopore Technologies (ONT). While SRS provides low-error, high depth sequencing, LRS has superior resolution and can thus delineate highly similar strains (Govender et al. 2021, Ren et al. 2021, Khedher et al. 2022, Chen et al. 2023). In most cases, a hybrid sequencing approach is used to achieve error-free and high-resolution strain typing (Lipworth et al. 2022, Meslier et al. 2022). However, this cannot be used in clinical application due to high TAT, high costs, and low reproducibility.

While operational taxonomic units suffice the interrogation of communities, strain-level resolution is imperative to differentiate pathogens and commensals as needed for clinical application. Ultimately, the choice of sequencing platform, depth of sequencing, bioinformatic strategy, and database diversity need to converge to provide the appropriate resolution of taxonomy, which determines the utility of NGS (Chiu and Miller 2019). Several metagenomic tools and databases have been developed to enable a variety of NGS applications for the detection and characterization of pathogens, enabling the use of genome sequencing as the final frontier for classification of pathogens (Grumaz et al. 2019, Ye et al. 2019, Avershina et al. 2022). Here, we explored third-generation LRS for its utility in supporting clinical decisions (Ren et al. 2021, Fu et al. 2022, Yonkus et al. 2022). The aim is to describe a workflow approach to simultaneously detect bacteria, fungi, and antibiotic resistance genes (ARGs) using amplicon-based LRS. We have identified several critical challenges, such as high error rates in sequencing, barcode mis-annotation, demultiplexing errors, cross-priming, the requirement of minimum sample DNA and library concentration to be resolved for the operational implementation of LRS in healthcare settings. By optimizing the minimum sample to DNA ratio, PCR conditions, sequencing protocols, adjusting post-sequencing filters and databases, and automating bioinformatics, we demonstrate a significant improvement in the reproducible identification of pathogens in clinical samples.

Methods

Samples

Twelve known ATCC/quality control laboratory organisms were used for the proof of concept and validation testing (Table 1). A metagenomic standard containing mixed genomic material of 11 strains (MSA-4000™) was also evaluated using an initial concentration of 1 ng μl^{-1} . Blinded evaluation of 114 clinical isolates of bacteria and fungi was performed in two tertiary care hospitals and one diagnostic laboratory (48, 18, and 48 samples, respectively). These clinical isolates were previously identified using automated or manual microbial culture testing; the identification of 18 of these isolates from one tertiary care hospital was reconfirmed with Sanger sequencing. Spiked human plasma experiments were performed with *Escherichia coli* ATCC 43895 (Table 1). Before testing, isolates were grown overnight in nutrient broth and spiked in the range 10^1 – 10^5 CFU ml^{-1} in 1 ml of plasma.

Laboratory isolates were maintained on Tryptone Soy Agar or Sabouraud's agar with monthly subculture performed to assess viability. Blood samples collected from patients with suspected or confirmed bloodstream infections after obtaining patient consent were evaluated ($n = 33$).

Sample processing

DNA was extracted from 1 ml of broth containing an overnight grown culture or 1 ml of spiked plasma containing 10^1 – 10^5 CFU ml^{-1} of organism that was centrifuged at $15\,000 \times g$ for 10 min. For clinical samples, plasma was separated from 8 ml ethylenediamine tetraacetic acid (EDTA) whole blood and centrifuged at $15\,000 \times g$ for 10 min. The pellet was subjected to DNA extraction as per the manufacturer's instructions (QIAmp DNA Mini Kit, 51304, Qiagen).

NGS library

The extracted DNA was normalized to 1 ng μl^{-1} (samples DNA < 1 ng μl^{-1} were not diluted) and enriched for microbial DNA targets (supplementary 1). The targets include the 16S rRNA gene for bacteria, 18S rRNA gene or internal transcribed spacer (ITS) for fungi, and ARGs include *bla*OXA type, *bla*CTX-M, *bla*SHV, *bla*TEM, *bla*NDM, *mecA*, *vanA*, *mcr*-1, and *aph* (3') (Murakami et al. 1991, Heuer et al. 1997, Lartigue et al. 2004, d'Azevedo et al. 2009, Dallenne et al. 2010, Yu et al. 2013, Mushi et al. 2014, Wang et al. 2014, Ogutu et al. 2015, Yang et al. 2019). These targets were selected to cover clinically relevant ARGs that confer resistance to primary antibiotic classes such as cephalosporins, carbapenams, methicillin, colistin, and vancomycin. A PCR mix consisting of 18 μl of master mix and 2 μl of the template DNA per reaction was set up. The thermal cycler programme of initial denaturation at 95°C for 5 min, followed by 25 cycles of 30 s at 95°C, 30 s at 62°C, and 120 s at 72°C, with a final extension of 300 s at 72°C (Q5 High-Fidelity PCR Kit, M0492, New England Biolabs) was set up.

The library was prepared with AMPure XP bead purified amplicons (A63882, Beckman Coulter Life Sciences). End preparation was performed using a reaction mix containing 4.25 μl of the end preparation reaction buffer, 0.75 μl of end preparation enzyme mix (E7645, New England Biolabs), and 10 μl of amplicon. The mix was incubated at 20°C for 15 min, followed by 65°C for 15 min. DNA barcodes (EXP-NBD196, ONT) were attached using 3 μl of end prepared contents, 5 μl of blunt/TA ligase master mix (M0367S, New England Biolabs), and 2 μl of unique barcodes. The mix was incubated at 20°C for 15 min followed by 65°C for 15 min. Post barcode ligation, 10 μl of each sample was pooled and purified using 1.5 X of AMPure XP beads followed by a wash with Short Fragment Buffer (SFB) and a subsequent wash with 80% ethanol. The library was eluted in 32 μl nuclease free water. Adapters (EXP-AMII001, ONT) were then attached by adding 5 μl adapters to 30 μl of elute and 5 μl of Quick T4 DNA ligase and 10 μl of quick ligation buffer (E6056, New England Biolabs). This was incubated at room temperature for 15 min. A final purification was performed with 0.4X of AMPure XP beads with two washes with 200 μl of SFB buffer (ONT). The library was eluted in 15 μl elution buffer.

Table 1. List of the standard samples used in the study.

S. No.	Sample	Sample source
1	<i>Escherichia coli</i> (positive control)	ATCC 43895
2	Nuclease free water (negative control)	Sigma Aldrich Catalogue No. R0581
3	<i>Burkholderia pyrrhotina</i>	Culture isolate confirmed through WGS
4	<i>Enterococcus faecalis</i>	ATCC 29212
5	<i>Enterococcus faecium</i>	Culture isolate confirmed through WGS
6	<i>Aspergillus fumigatus</i>	Sigma Aldrich Catalogue No. MB00038
7	<i>Acinetobacter baumannii</i>	Culture isolate confirmed through WGS
8	<i>Mycobacterium tuberculosis</i>	ATCC 25618D-2
9	<i>Pseudomonas aeruginosa</i>	ATCC 27853
10	<i>Escherichia coli</i> (colistin-resistant)	Culture isolate confirmed through WGS
11	<i>Candida tropicalis</i>	ATCC 66029
12	Metagenomic control material for pathogen detection	ATCC MSA-400™
13	<i>Escherichia coli</i> (extended-spectrum beta-lactamase producer)	Culture isolate confirmed through WGS
14	Spiked plasma samples ($n = 24$)	Normal human plasma spiked with <i>E. coli</i> ATCC 43895
15	Blinded culture isolates ($n = 114$)	Clinical isolates confirmed through standard culture testing
16	Blood samples ($n = 33$)	From patients with suspected bloodstream infection

Nanopore sequencing

The library was prepared for ONT sequencing in accordance with the manufacturer's instructions (SQL-LSK109, ONT). The R9.4.1 MinION flow cells and Flongle flow cells were used as per the number of samples to be sequenced. The flow cells were primed as per manufacturer's instructions, followed by loading of the library. Sequencing was carried out over 6–48 h as required. Various different MinKNOW versions (March 2022–October 2023) that were released during the study time period were used for analysis.

Analysis

For analysis, reads with Q -value >10 , which corresponds to an estimated $>98\%$ of reads to be below the 10% error rate (Delahaye and Nicolas 2021), were obtained in the FASTQ format after basecalling and demultiplexing using dual barcodes. The host contamination was reduced by removing reads mapping to the host genome version (GRCh38) using minimap2 version 2.24-r1122 (Li 2018) and samtools (Danecek et al. 2021). The reads were subjected to multiple filtering processes, i.e. the chimaera and adapters were removed using the Porechop v.0.2.4. Reads with low quality score and unwanted length were removed using NanoFilt v2.8.0 (De Coster et al. 2018). All the file conversions were performed using seqtk 1.3-r106. The reads were screened using several tools (supplementary 2) and custom (supplementary 3) and curated databases (supplementary 2) to identify the relevant pathogen and ARGs. The reads were screened using BLAST v.2.12.0 (Altschul et al. 1990) to identify the relevant pathogen, whereas ARGs were identified using staramr (v.0.7.2), which uses the ResFinder (updated: 16 July 2018) database (Bharat et al. 2022). An automated bioinformatic analysis tool developed using in-house scripts, generates alignments of reads to a curated database and identifies the presence of bacterial and fungal pathogens, and ARGs. This tool was being developed to detect all bacteria and fungi and several classes of antibiotics, including but not limited to extended-spectrum β -lactamases, carbapenem resistance-conferring genes, and vancomycin resistance-conferring genes. Figure 1 indicates the analytical steps undertaken to determine the presence of pathogen or ARG in each barcoded sample.

Results

Identification of pathogens using amplicon-based LRS

The proof of concept testing for the identification of organisms using amplicon-based LRS with ONT was demonstrated using the 16S rRNA and 18S rRNA genes as targets to identify bacteria and fungi, respectively (Heuer et al. 1997, Yu et al. 2013). Sequencing was performed on a Flongle flow-cell over 6 h on representative bacterial (*E. coli*) and fungal samples (*Candida tropicalis*), which could be successfully identified. A majority of reads matched to the relevant pathogen with 80.54% of reads matching the correct genus and species in case of bacteria and 97.15% of reads matching the correct genus and species in case of fungi (Table 2). The reads that were not annotated to *E. coli* or *C. tropicalis* contained mis-annotated reads that matched closely related organisms within the Enterobacteriales family for *E. coli* and within the same genera for *C. tropicalis*, respectively. The proof of concept was additionally verified using standard metagenomic material. We successfully identified all 10 organisms present in the metagenomic standard (supplementary 4). The next step was to ascertain the limit of detection (LOD). For this purpose, a laboratory culture of *E. coli* was serially diluted to contain 10^1 – 10^5 CFU ml $^{-1}$ of the organism. Each culture and dilution were spiked in plasma and tested as individual samples to identify LOD. The LOD was found to be 10^3 CFU ml $^{-1}$ for *E. coli* (Table 3).

The possibility of expanding such an assay to identify both pathogens as well as ARGs was explored by testing multiple target genes, including 16S rRNA for bacteria, 18S rRNA and ITS (Heuer et al. 1997, Yu et al. 2013, Wang et al. 2014) for fungi as well as several ARGs covering extended-spectrum beta-lactamases (ESBLs; *bla*OXA type, *bla*CTX-M, *bla*SHV, *bla*TEM), carbapenemases (*bla*NDM), aminoglycosides (*aph* (3')), colistin (*mcr*-1), vancomycin (*vanA*), and methicillin (*mecA*). These targets were tested using laboratory cultures confirmed to contain the relevant ARGs using whole genome sequencing. In each case, the target gene was successfully identified (Table 4).

The ability of the assay to identify a wide range of bacteria and fungi using the 16S rRNA gene and the 18S rRNA/ITS regions, respectively, was evaluated through blinded testing of previously identified clinical isolates. A concordance with

Table 2. Identification of bacteria and fungi using amplicon-based LRS.

S. no.	Target	Sample source	Organism detected	Number of reads matching target	Proportion of reads matching pathogen
1	16S rRNA	Culture	<i>Escherichia coli</i>	3378	80.54%
2	18S rRNA	Culture	<i>Candida tropicalis</i>	1566	97.15%

Table 3. LOD of identifying *E. coli* in spiked plasma using 16S rRNA.

S. no.	Expected pathogen	Identified pathogen	Number of reads matching target	Proportion of reads matching pathogen
1	<i>Escherichia coli</i> 10 ⁵ CFU ml ⁻¹	<i>Escherichia coli</i>	1339	72.89
2	<i>Escherichia coli</i> 10 ⁴ CFU ml ⁻¹	<i>Escherichia coli</i>	575	74.1
3	<i>Escherichia coli</i> 10 ³ CFU ml ⁻¹	<i>Escherichia coli</i>	185	62.93
4	<i>Escherichia coli</i> 10 ² CFU ml ⁻¹	Not detected	0	0
5	<i>Escherichia coli</i> 10 ¹ CFU ml ⁻¹	Not detected	0	0
7	Non-spiked plasma (control)	Not detected	0	0

Table 4. Identification of pathogenic organisms and ARGs using specific gene targets.

S. no.	Expected target	Expected organism/ARG	Target identified	No. of reads matching target
1	16S	<i>Escherichia coli</i>	<i>Escherichia coli</i>	329
2	ITS	<i>Candida tropicalis</i>	<i>Candida tropicalis</i>	870
3	18S	<i>Candida tropicalis</i>	<i>Candida tropicalis</i>	1566
4	<i>aph</i> (3')	<i>aph</i> (3')	<i>aph</i> (3')	605
5	<i>bla</i> CTXM-15	<i>bla</i> CTXM-15	<i>bla</i> CTXM-15	662
6	<i>mecA</i>	<i>mecA</i>	<i>mecA</i>	552
7	<i>bla</i> NDM	<i>bla</i> NDM	<i>bla</i> NDM	639
8	<i>bla</i> SHV	<i>bla</i> SHV	<i>bla</i> SHV	14
9	<i>bla</i> TEM	<i>bla</i> TEM	<i>bla</i> TEM	600
10	<i>vanA</i>	<i>vanA</i>	<i>vanA</i>	71
11	<i>bla</i> OXA	<i>bla</i> OXA	<i>bla</i> OXA	565

kappa (k) = 0.83 was observed for a species level match and k = 0.90 for a genus level match (Table 5). Organisms that could not be identified were either not called owing to low reads obtained in the same or the pathogen was not included as part of the curated database (supplementary 2 and 3).

In order to simplify the workflow, these targets were multiplexed into a single assay to simultaneously identify bacteria, fungi, and ARGs. Gene targets for the same included 16S rRNA gene, 18S rRNA gene/ITS region, *bla*OXA, *bla*CTX-M, *bla*SHV, *bla*TEM, *bla*NDM, *vanA*, *mecA*, *aph* (3'). This was tested using culture samples as well as blood samples from patients with suspected bloodstream infections. A multiplexed format combined with real-time sequencing could consistently identify bacteria, fungi, and ARGs within 12 h as a single test solution (Fig. 2). Several bacterial and fungal species were identified along with ARGs in some samples (Table 6). In a multiplexed format, some challenges became apparent, such as 40%–98% quality filtered reads of each sample were not being assigned to a target. Another challenge faced was the identification of ARGs in a plasma sample but not the organism.

Initial challenges in multiplexed amplicon-based identification of targets using ONT sequencing

The workflow described in Fig. 1 raised a few aspects that required further consideration. The targets detected by varied tools and databases differed in their consistency of identification or time taken to run the analysis (supplementary 2). When tested in triplicate, samples containing multiple targets consistently had 80%–85% of reads that would pass other qual-

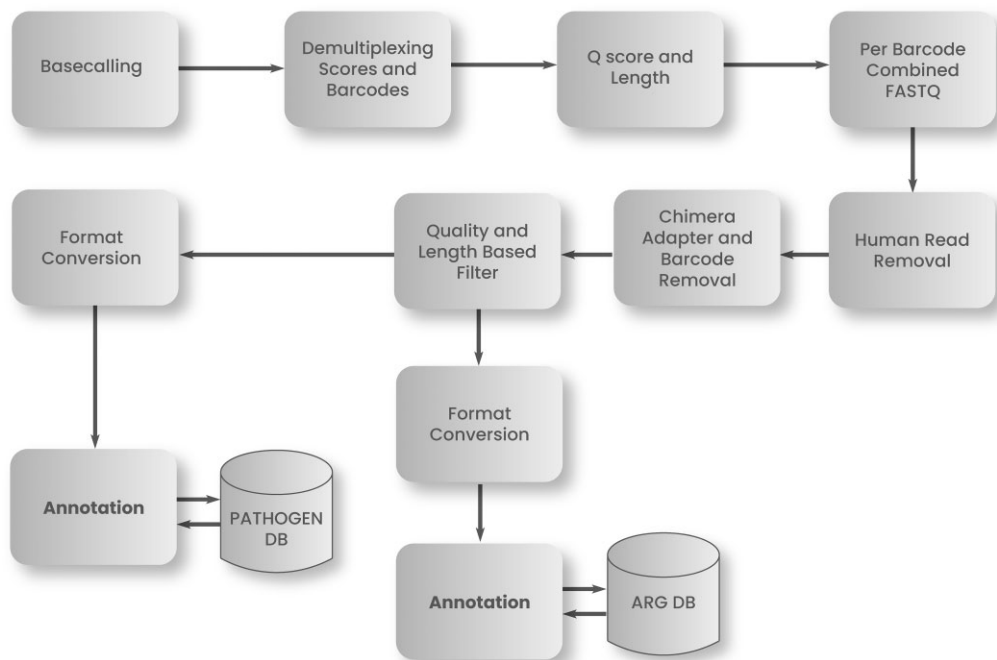
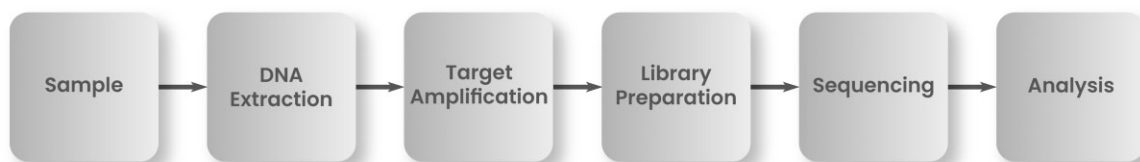
ity criteria; however, the reads would not match the relevant target (Table 7). A potential reason for the non-annotation of amplicon-based reads into pathogen or ARG was identified to be non-specific amplification by the 18S rRNA and *bla*CTX-M target primers (Wang et al. 2014, Ogutu et al. 2015). The 18S rRNA target identified fungi from culture but resulted in non-specific amplification of human target regions when tested with blood samples. The *bla*CTX-M primers non-specifically amplified targets in the human genome. Table 8 demonstrates that despite an adequate throughput, <5% of reads could be annotated into the relevant targets; Table 9 shows resolution of this by replacing the 18S rRNA gene target with ITS for identification of fungi and alternative *bla*CTX-M primers for the same set of samples (Lartigue et al. 2004). The improvement was seen in both culture as well as blood samples.

In the case of a multiplexed workflow, quality filters and threshold-based cutoffs are required (Table 10). Low numbers of unexpected reads have been observed in samples, such as the presence of bacterial plasmid-based ARGs in confirmed fungal samples (*vanA* seen in *Aspergillus fumigatus*) or the presence of ARGs commonly seen in Gram-negative samples (*bla*NDM, *bla*TEM, *bla*SHV) being present in Gram-positive (*Enterococcus faecium*) samples. In order to better understand whether the unexpected reads were observed at a lower stringency of demultiplexing, samples were demultiplexed at accuracy levels. Demultiplexing at an accuracy of >95% accurately identified the presence or absence of an organism as expected. In this case, the two blood samples were expected to be negative, and the positive control expected to contain *E. coli* (Table 11).

Table 5. Evaluating the concordance of the assay using previously identified clinical isolates ($n = 114$).

Location	Concordant	Discordant	Observed agreement (K)
Tertiary care hospital 1 ($n = 48$)	38	10	0.79
Tertiary care hospital 2 ($n = 18$)	14	4	0.77
Diagnostic laboratory 1 ($n = 48$)	42	6	0.88
Overall	94	20	0.83

Note: Tests at Tertiary hospital 1 and diagnostic laboratory were performed using 16S rRNA and 18S rRNA genes as targets. Tests at Tertiary hospital 2 were performed using 16S rRNA and ITS as targets (supplementary 5). An additional eight samples matched at the genus level ($K = 0.90$).

**Figure 1.** Bioinformatic workflow for the simultaneous detection of bacteria, fungi, and ARGs. DB: Database; ARG: Antibiotic resistance gene.**Figure 2.** Assay workflow for the simultaneous detection of bacteria, fungi, and ARGs.**Table 6.** Multiplexed amplicon-based simultaneous identification of bacteria, fungi, and ARGs in pure isolates ($n = 5$) and blood samples ($n = 2$).

S. no.	Sample source	Total reads	Quality filtered reads	Organism detected	No. of reads matching organism	<i>aph(3')</i>	<i>blaNDM</i>	<i>blaTEM</i>	<i>vanA</i>	<i>mcr-1</i>
1	DNA	5045	3516	<i>Burkholderia stabilis</i>	2103	-	-	-	-	-
2	DNA	10 611	6987	<i>Escherichia coli</i>	2542	618	-	-	-	544
3	DNA	3396	1841	<i>Enterococcus faecium</i>	1698	155	-	-	274	-
4	DNA	5422	2313	<i>Pseudomonas aeruginosa</i>	684	-	-	-	-	-
5	DNA	26 498	19 881	<i>Enterococcus faecalis</i>	1916	154	-	-	-	-
6	DNA	29 518	21 165	<i>Aspergillus fumigatus</i>	500	-	-	-	-	-
7	Blood	185	47	<i>Acinetobacter baumannii</i>	27	-	-	-	-	-
8	Blood	336	170	Not detected	-	-	36	37	-	-

Note: Genes targeted in the multiplexed assay included 16S rRNA, ITS, *blaOXA*, *aph(3')*, *blaCTX-M*, *blaNDM*, *mecA*, *blaSHV*, *blaTEM*, *vanA*, and *mcr-1*.

Table 7. Distribution of reads in multiplexed amplicons targeting organisms or ARGs.

S. no.	Organism present	ARG present	Total reads	Quality filtered reads	No. of reads matching organism	<i>aph(3')</i>	<i>vanA</i>	<i>mcr-1</i>	Proportion of annotated reads
1	<i>Enterococcus faecium</i>	<i>VanA</i> , <i>aph(3')</i>	9733	7549	1182	26	186	-	19.35%
2	<i>Enterococcus faecalis</i>	NA	3129	2024	1881	-	-	-	99%
3	<i>Escherichia coli</i>	<i>mcr-1</i> , <i>aph(3')</i>	15 582	11 327	491	582	-	556	14.01%
4	<i>Escherichia coli</i>	NA	1310	942	790	-	-	-	81.85%

Proportion of annotated reads is calculated as: (sum of reads annotated to any target/quality filtered reads) $\times$ 100. As indicated by the values in bold text, the proportion of all quality filtered reads matching a target (pathogen or ARG) is significantly lower in cases where multiple targets are present in the same samples, i.e. a pathogen and an ARG. This was tested in triplicate with similar results.

Table 8. Identification of fungus and *bla*CTX-M using 18 S as a target and set 1 primers for *bla*CTX-M.

S. no.	Sample ID	Target	Sample type	Total reads	Quality filtered reads	Organism present	No. of reads matching organism	ARG present	No. of reads matching ARG
1	HAPL_ID23_002	18S	Culture	468 493	8308	<i>Candida albicans</i>	6891	Not detected	Not detected
2	HAPL_ID23_020	<i>bla</i> CTX-M	Culture	11 702	10 221	Not detected	Not detected	<i>bla</i> CTX-M	662
3	HAPL_ID23_040	16S, 18S, and <i>bla</i> CTX-M	Blood	3843	7	Not detected	Not detected	Not detected	Not detected

Table 9. Identification of fungus and *bla*CTX-M using ITS as a target and set 2 primers for *bla*CTX-M.

S. no.	Sample ID	Target	Sample type	Total reads	Quality filtered reads	Organism present	No. of reads matching organism	ARG present	No. of reads matching ARG
1	HAPL_ID23_041	ITS and 16S	Culture	4133	1728	<i>Candida tropicalis</i>	1566	ND	Not detected
2	HAPL_ID23_042	<i>bla</i> CTX-M	Culture	322	267	Not detected	Not detected	<i>bla</i> CTX-M	232
3	HAPL_ID23_043	16S, ITS, and <i>bla</i> CTX-M	Blood	947	136	<i>Escherichia coli</i> ; <i>Malassezia</i> spp.	2911	Not detected	Not detected

The set of samples was tested using a curated custom database as well as a larger curated database (supplementary 5). However, a database limited in terms of sequence redundancy and species heterogeneity affected the specificity. The result was that samples containing an organism would be categorized as a false negative (Table 12). Alternatively, organisms would match to a closely related organism. Using an expanded database could overcome such an issue (Table 12; Curry et al. 2022).

Establishing a workflow with potential clinical application

A genomics-based approach to detecting infections has the potential to be a one-test solution to identify several pathogens simultaneously. The clinical viability of such a workflow should consider the sensitivity and specificity of identification of the pathogen. An amplicon-based approach can improve sensitivity by selectively enriching microbial targets and can be used to test samples directly without the need for culture. A combination of LRS and automated analysis can reduce testing times to a few hours. The workflow described here could reproducibly identify pathogens and ARGs in a single assay. Further, the addition of cutoffs to identify the majority of reads matching to an organism could offset sequencing related errors. We found the use of high-accuracy basecalling to improve specificity, alternatively, using fast-basecalling with Q -value > 10 combined with double-sided barcoding also improved specificity. A BLAST-based threshold e -value 0.001 and $>80\%$ identity was used to identify the best hit for a read when screened against the database (Altschul et al.

1990). An organism was confidently identified when $\geq 60\%$ reads matched the given organism with a minimum of 10 reads showing a match to that particular organism. ARGs were identified using staramr (v.0.7.2), wherein reads with $>90\%$ identity and $>60\%$ overlap of the gene (Bharat et al. 2022) were classified when >10 reads matched to the same ARG. We observed that the use of an expanded database had greater utility in identifying a varied spectrum of pathogens and minimized no identification or misclassification of a pathogen (Table 12). This strategy, when evaluated in 28 blood samples taken from patients with suspected bloodstream infections, could reproducibly and correctly identify pathogens. The 28 samples were tested using 2 sequencing runs that included 14 samples each, a positive (*E. coli* culture) and negative control (nuclease free water). This testing could identify relevant pathogens and ARGs (Table 13). A pathogen clinically correlating with the patients' symptoms could be identified in 4 of the 28 samples including 1 case where a pathogen and ARG could be simultaneously identified. The total workflow of processing and analysing the clinical samples could be achieved within a turn around time of 12 h.

Discussion

The utility of genome sequencing as a diagnostic tool is gaining importance. Several applications are available today that bring us closer to our goal of using precision medicine or tailor-made regimens in treating patients appropriately. In the case of infectious diseases, genome sequencing has already superseded culture or conventional methods for antimicrobial

Table 10. Profile of annotated reads among known samples.

S. no.	Sample set	Total reads (x1000)	Passed reads	Filtered reads	Organism detected	Organism reads	blaOXA	aph(3')	blaNDM	mecA	blaSHV	blaTEM	blaCTX-M	vanA	mer-1
1	Sample set 1	372.14	9269	7125	<i>Burkholderia stabilis</i>	4349	-	2	-	-	-	12	-	-	-
2			39904	35444	<i>Escherichia coli</i>	4128	-	654	-	-	-	17	-	-	542
3			14306	12563	<i>Enterococcus faecium</i>	3245	-	168	4	-	3	3	-	493	-
4			3315	2761	<i>Pseudomonas aeruginosa</i>	2353	-	2	-	-	-	-	-	-	-
5			6622	5628	<i>Enterococcus faecalis</i>	5320	-	312	-	-	-	-	-	-	-
6			3410	1307	<i>Aspergillus fumigatus</i>	1022	-	21	-	-	-	-	-	1	-
7			14209	11940	<i>Escherichia coli</i>	6992	-	11	-	-	-	-	-	2	1
8			195	124	Nuclease free water	-	-	89	-	-	-	13	-	2	2
1	Sample set 2	266.72	832	208	<i>Burkholderia stabilis</i>	118	-	-	-	-	-	-	-	-	-
2			11714	5162	<i>Escherichia coli</i>	295	-	583	-	-	-	-	-	-	516
3			3588	1353	<i>Enterococcus faecium</i>	294	1	12	2	-	-	-	-	4	-
4			415	158	<i>Pseudomonas aeruginosa</i>	123	-	-	-	-	-	-	-	-	-
5			972	285	<i>Enterococcus faecalis</i>	246	-	18	-	-	-	-	-	-	-
6			618	123	<i>Aspergillus fumigatus</i>	88	-	-	-	-	-	-	-	-	-
7			1952	711	<i>Escherichia coli</i>	350	-	-	-	-	-	-	-	-	-
8			42	2	Nuclease free water	-	-	1	-	-	-	-	-	-	-

Table 11. Detection of targets at varying accuracy thresholds of demultiplexing.

S. no	Demultiplexing accuracy (%)	Sample 1		Sample 2		Positive control	
		Pathogen detected	No. of reads matching organism	Pathogen detected	No. of reads matching organism	Pathogen detected	No. of reads matching organism
1	60	<i>Acinetobacter baumannii</i>	67	<i>Acinetobacter baumannii</i>	10	<i>Escherichia coli</i>	2386
2	70	<i>Acinetobacter baumannii</i>	58	<i>Acinetobacter baumannii</i>	10	<i>Escherichia coli</i>	2180
3	80	<i>Acinetobacter baumannii</i>	46	Not detected	-	<i>Escherichia coli</i>	1719
4	85	<i>Acinetobacter baumannii</i>	31	Not detected	-	<i>Escherichia coli</i>	1424
5	90	<i>Acinetobacter baumannii</i>	20	<i>Acinetobacter baumannii</i>	36	<i>Escherichia coli</i>	1199
6	95	Not detected	-	Not detected	-	<i>Escherichia coli</i>	575

Note: Both samples were expected to be negative, and the positive control to have *E. coli*.

Table 12. Identification of pathogens using relevant improvements of an expanded database and cut-off.

S. no.	Total reads	Quality filtered reads	Expected pathogen	Pathogen detected with custom curated DB and cutoff	Pathogen detected with custom curated DB and no cutoff	Pathogen detected with expanded DB and cutoff	Pathogen detected with expanded DB and no cutoff
1	9276	6245	<i>Shigella boydii</i>	<i>Escherichia coli</i>	<i>Escherichia coli</i>	Not detected	<i>Shigella boydii</i>
2	9269	7125	<i>Burkholderia pyrrrocinia</i>	Not detected	<i>Burkholderia cepacia</i>	<i>Burkholderia stabilis</i>	<i>Burkholderia stabilis</i>
3	1852	1123	<i>Shigella dysenteriae</i>	<i>Escherichia coli</i>	<i>Escherichia coli</i>	<i>Shigella flexneri</i>	<i>Shigella flexneri</i>
4	2639	1411	<i>Candida tropicalis</i>	<i>Candida tropicalis</i>	<i>Candida tropicalis</i>	<i>Candida tropicalis</i>	<i>Candida tropicalis</i>
5	6781	5438	<i>Acinetobacter baumannii</i>	<i>Acinetobacter baumannii</i>	<i>Acinetobacter baumannii</i>	<i>Acinetobacter baumannii</i>	<i>Acinetobacter baumannii</i>
6	905	644	<i>Shigella flexneri</i>	<i>Escherichia coli</i>	<i>Escherichia coli</i>	<i>Shigella flexneri</i>	<i>Shigella flexneri</i>
7	301	242	<i>Candida tropicalis</i>	Not detected	Not detected	<i>Candida africana</i>	<i>Candida africana</i>
8	599	273	<i>Brucella melitensis</i>	Not detected	<i>Stenotrophomonas maltophilia</i>	<i>Brucella melitensis</i>	<i>Brucella melitensis</i>
9	2895	1486	<i>Candida albicans</i>	Not detected	<i>Candida parapsilosis</i>	<i>Candida africana</i>	<i>Candida africana</i>
10	5798	3550	<i>Mycobacterium tuberculosis</i>	Not detected	<i>Salmonella bongori</i>	<i>Mycobacterium spp.</i>	<i>Mycobacterium canettii</i>

DB, Database; custom curated DB (supplementary 2); expanded DB (Currey et al. 2022).

susceptibility testing of diseases such as tuberculosis (Walker et al. 2013, Chatterjee et al. 2017, Street et al. 2022, Zade et al. 2022, CRYPTIC Consortium and the 100 000 Genomes Project et al. 2018). For the other infections, DNA sequencing of microorganisms from pure cultures or microbiomes has been successfully used to overcome the limitations of microbial cultures and has deepened our understanding of the diversity and functionality of these organisms and communities. The clinical application of microbial (Blauwkamp et al. 2019, Grumaz et al. 2019, Tact et al. 2020, Fu et al. 2022) genomics, however, remains limited, primarily suffering from a lack of standardization of methods of nucleic acid extraction, sequencing workflows, and non-robust bioinformatics (Petti 2007, Church et al. 2020). Real-time LRS has the potential to be implemented for the rapid detection of several pathogens simultaneously. While several studies have evaluated LRS on pure cultures, clinical isolates, or blood cultures, challenges remain in a direct-from-sample approach and setting up the same as a workflow that fits into routine healthcare as it stands today (Tact et al. 2020, Ren et al. 2021, Avershina et al. 2022).

Here we report several improvements in the end-to-end protocols for the use of ONT for simultaneous detection of bacteria, fungi, and ARGs using culture isolates and in blood samples of suspected cases of bacteremia and fungemia. An amplicon-based approach was taken to reduce the back-

ground of host nucleic acid as well as enrich the microbial targets of interest. Optimization of targets and primers, setting thresholds for the interpretation of quality filtered sequenced reads, careful curation of the reference database and concurrent use of multiple annotation algorithms, when combined, generate a reproducible and easy-to-use method for enabling the use of ONT as a diagnostic solution for detecting bacteria, fungi, and ARGs in the clinical samples (Fig. 1).

Challenges arose in the selection of targets such as fungal targets, where 18S rRNA, initially selected owing to its potential of utility across a wider taxonomic range (Hugerth et al. 2014), cross-primed and amplified human DNA regions as well; shifting to ITS as a target resolved this issue as well as improved species level calling for fungi such as *Candida* sp. (Yu et al. 2013, Lücking et al. 2020). The selection of other targets and primers in a multiplexed assay is also important as seen with the *bla*CTX-M primer that resulted in non-specific target amplification and reduced specificity of calling this ARG. This was resolved by using an alternative set of primers (Lartigue et al. 2004, Ogutu et al. 2015). Although the library was well standardized, sample-to-sample variability and process controls need to be maintained. This was accomplished by maintaining at least one known positive sample in the sequencing batch and loading libraries containing >10 ng of DNA. The high error rates of ONT sequencing were offset by using a higher stringency of de-multiplexing to avoid misclas-

Table 13. Findings from testing of clinical samples using the assay.

S. no.	Sample ID	Total reads	Quality filtered reads	Pathogen detected	ARG detected
1	HAPL_ID23_088	246	25	<i>Escherichia coli</i>	Not detected
2	HAPL_ID23_099	99	8	Not detected	Not detected
3	HAPL_ID23_093	96	8	Not detected	Not detected
4	HAPL_ID23_090	165	9	Not detected	Not detected
5	HAPL_ID23_100	154	43	Not detected	Not detected
6	HAPL_ID23_097	105	21	Not detected	Not detected
7	HAPL_ID23_089	77	17	Not detected	Not detected
8	HAPL_ID23_092	190	27	Not detected	Not detected
9	HAPL_ID23_103	149	15	Not detected	Not detected
10	HAPL_ID23_094	410	57	<i>Escherichia coli</i> ^a	<i>bla</i> NDM
11	HAPL_ID23_095	89	20	Not detected	Not detected
12	HAPL_ID23_098	183	29	<i>Escherichia coli</i>	Not detected
13	HAPL_ID23_096	118	9	Not detected	Not detected
14	HAPL_ID23_091	251	27	Not detected	Not detected
15	HAPL_ID23_102	4308	3104	<i>Escherichia coli</i> (positive control)	Not detected
16	HAPL_ID23_101	107	13	Not detected (negative control)	Not detected
17	HAPL_ID23_112	902	30	<i>Enterococcus faecalis</i>	Not detected
18	HAPL_ID23_106	1869	10	Not detected	Not detected
19	HAPL_ID23_117	570	6	Not detected	Not detected
20	HAPL_ID23_105	208	2	Not detected	Not detected
21	HAPL_ID23_115	278	7	Not detected	Not detected
22	HAPL_ID23_118	56	0	Not detected	Not detected
23	HAPL_ID23_119	1218	16	Not detected	Not detected
24	HAPL_ID23_111	400	5	Not detected	Not detected
25	HAPL_ID23_116	1377	6	Not detected	Not detected
26	HAPL_ID23_114	14	0	Not detected	Not detected
27	HAPL_ID23_110	5	0	Not detected	Not detected
28	HAPL_ID23_104	2088	3	Not detected	Not detected
29	HAPL_ID23_113	327	0	Not detected	Not detected
30	HAPL_ID23_109	99	4	Not detected	Not detected
31	HAPL_ID23_108	5983	3645	<i>Escherichia coli</i> (positive control)	Not detected
32	HAPL_ID23_107	6	0	Not detected (negative control)	Not detected

Positive control used was *E. coli* (ATCC 43895). Negative control used was nuclease free water. Serial numbers 1–16 and 17–32 were run as two independent sequencing runs.

^aA majority of reads aligned with *E. coli* but were <60% of all quality filtered reads.

sification of reads within a sequencing batch, a problem that was exacerbated by the presence of multiple targets within the same sample, such as a pathogen and an ARG target (Table 6). In addition to the increased stringency, we established the requirement of a minimum threshold for annotated reads for the correct interpretation of the sample. The ultimate goal was to develop a fully automated bioinformatic software that would provide clinically actionable reports for the use by healthcare providers with little or no expertise in bioinformatics. We employed a bioinformatic strategy that used both global and local alignment to generate a count of best hits and then applied a threshold to report the predominant microbial species in the sample (Altschul et al. 1990, Carattoli et al. 2014, Bharat et al. 2022). This implied that the target enrichment had to be improved to reduce artefacts available for sequencing and a curated database that prevented the mis-annotation of reads as was seen with using a redundant reference database (Sayers et al. 2021, Curry et al. 2022, Feldgarden et al. 2022).

NGS-based technologies have the potential to overcome several limitations of current day diagnostics. Studies have already established an improved sensitivity and specificity compared with standard-of-care tests. Current methods of detection lag behind the infection and are prescribed in reaction to clinical observations (Grumaz et al. 2019, Vincent et al. 2020, Fu et al. 2022). Thus, owing to a lack of antimicrobial and diagnostic stewardship, general clinical practice tends

to use antimicrobial agents indiscriminately during clinical events (infectious syndromes) or procedures (major surgery). This rampant abuse of antibiotics compromises antimicrobial stewardship programmes, and is pushing the spectrum of antimicrobial resistance across the globe. Highly sensitive methods of pathogen screening are the need of the hour for patients admitted to tertiary care, more so for patients in intensive care (WHO 2022). Reliable, on-site, screening of pathogens in tertiary care centres can enable healthcare providers to stay ahead of the infection. Here, we explored ONT sequencing as an affordable screening platform for pathogen screening; wherein complex diagnostics were implemented in a simpler form that could be driven by non-experts.

Conclusion

The introduction of NGS technologies has ushered in new possibilities in the domain of clinical microbiology and infectious disease management. Its use in infectious disease and particularly in critically ill patients has been widely explored. The need for genomic methods of pathogen screening in healthcare settings is exacerbated due to the rapid rise of drug resistance and related outbreaks of nosocomial infections (Vincent et al. 2020). Such an approach holds promise as a potential single assay to identify any pathogen. Direct-from-sample, on-site sequencing combined with automated bioinformatics is the need of the hour especially in critical care settings. In addition to en-

abling rapid detection of infection, a wider availability of such NGS solutions will enable better clinical annotation of genetic sequences leading to increased specificity of clinical pathogen genomics. Ultimately, the increased use of NGS in clinical microbiology will provide the necessary quality assured molecular diagnostic tools that can then set the standards for global adoption. The study here, performed across multiple tertiary care settings in India is a first step towards the same.

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Supplementary data

Supplementary data is available at *JAMBIO Journal* online.

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Resources, Supervision, Writing – original draft, Writing – review & editing)

Data availability

The datasets presented in this study can be found in NCBI repositories under Bioproject PRJNA1074527 and Bioproject PRJNA1074529.

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